Eon	Era	Period	Epoch	Outline	Events	Beginning
Hadean				Formation of the Earth and Moon. Likely emergence of the last universal common ancestor (LUCA). Atmospheric oxygen exists only in trace amounts.	— **4.576 ± 0.16 Ga:** Formation of Earth. — **4.4 Ga:** Theia, a Mars-sized planetary body, impacts Earth, leading to the formation of the moon. — **4.3 Ga:** Last potentially planet-sterilizing extraterrestrial impact. — **4.04 Ga:** Formation of the oldest minerals currently dated on Earth, in the form of zircon crystals. — **4.01 Ga:** Oldest potentially biogenic carbon, found preserved in zircon.	4.57 Ga
Archean	Eoarchean			Increasing formation of continental crust. Appearance of anoxygenic photosynthesis.	— **4.031 Ga:** Formation of the oldest known intact rock formations on Earth. — **3.7 ± 0.1 Ga:** Banded iron formations (BIFs) suggest presence of anoxygenic photosynthesizers.	4.03 Ga
	Paleoarchean			Microbial mats become common.	— **3.48 Ga:** Microbially induced sedimentary structures (MISS) give strong evidence for the presence of microbial mats. — **3.33 Ga:** Presence of directly preserved microbial mats.	3.6 Ga
	Mesoarchean				— **3.2 Ga:** Microbial life settles in rivers. — **≈ 3 Ga:** Possible onset of plate tectonics. — **2.9 Ga:** Oldest evidence of glaciation. — **2.8 ± 0.4 Ga:** Emergence of cyanobacteria, the first oxygenetic photosynthesisers.	3.2 Ga
	Neoarchean			Major diversification of cyanobacteria, including multicellularity. Proliferation of oxygen-producing life gradually increases the amount of atmospheric oxygen.		2.8 Ga
Proterozoic	Paleoproterozoic	Siderian		The great oxidation event causes the extinction of many early anaerobic organisms. A day is 16.9 ± 0.2 hours long.	— **2.44 Ga:** Great Oxidation Event.	2.5 Ga
		Rhyacian		Increase of atmospheric oxygen leads to a major expansion of life, enabled by the efficiency of aerobic metabolisms.		2.3 Ga
		Orosirian			— **2.0 ± 0.3 Ga:** Eukaryogenesis! A possible early candidate is *Grypania*, a tube-shaped organism with a size over one centimeter. — **1.9 ± 0.1 Ga:** Formation of the Columbian supercontinent.	2.05 Ga
		Statherian				1.8 Ga
	Mesoproterozoic	Calymmian			— **1.6 Ga:** Presumed appearance of eukaryotic red algae!	1.6 Ga
		Ecstasian			— **1.4 ± 0.1 Ga:** Fragmentation of the Columbian supercontinent.	1.4 Ga
		Stenian			— **1.2 Ga:** Collision between the continents of Laurentia and Amazonia marks the first step in the assembly of the Rodinian supercontinent.	1.2 Ga
	Neoproterozoic	Tonian			— **725 ± 75 Ma:** East African orogeny marks the beginning of the formation of Gondwana.	1 Ga
		Cryogenian		Fragmentation of the Rodinian supercontinent. Two likely Snowball Earth glaciations.	— **720 – 658 Ma:** Sturtian glaciation. — **655 – 635 Ma:** Marinoan glaciation.	720 Ma
		Ediacaran		Gondwana assembles. Rapid rise in eukaryote diversity, likely including the first animals.	— **579.88 ± 0.44 – 579.63 ± 0.15 Ma:** Gaskiers glaciation.	635 Ma
Phanerozoic	Paleozoic	Cambrian	Terreneuvian	First recognizable members of modern animal phyla appear, including arthropods, molluscs, annelids, echinoderms, bryozoans, priapulids, brachiopods, hemichordates, and chordates.		539 Ma
			Series 2			521 Ma
			Miaolingian			507 Ma
			Firongian			497 Ma
		Ordovician	Lower	Earliest Fish appear. Early land plants spread.		487 Ma
			Middle			471 Ma
			Upper	Late Ordovician mass extinction leads to the loss of ≈ 85% of marine invertebrate species.		458 Ma
		Silurian	Llandovery	Major radiation of graptolites, bivalves, gastropods, nautiloids, brachiopods, and crinoids. Earliest jawed fish appear. Earliest vascular plants appear. Earliest land animals appear. Earliest freshwater fish appear.		443 Ma
			Wenlock			433 Ma
			Ludlow			427 Ma
			Pridoli			423 Ma
		Devonian	Lower	Vascular plants increase in size and develop large root systems. Jawed fish differentiate into ray-finned, lobe-finned, and cartilaginous fish. Tetrapods appear. Early amphibians move to land. Ammonoids appear.		420 Ma
			Middle			393 Ma
			Upper		— **372 Ma:** Late Devonian/Kellwasser extinction - loss of ≈40% of marine species. — **359 Ma:** End-Devonian/Hangenberg extinction.	382 Ma
		Carboniferous	Mississippian		— **335 Ma:** Gondwana, Euramerica and Sibera come together, forming Pangaea.	359 Ma
			Pennsylvanian			323 Ma
		Permian	Cisuralian			299 Ma
			Guadalupian			274 Ma
			Lopingian			260 Ma
	Mesozoic	Triassic	Lower			252 Ma
			Middle			247 Ma
			Upper			237 Ma
		Jurassic	Lower			201 Ma
			Middle			174 Ma
			Upper			161 Ma
		Cretaceous	Lower			143 Ma
			Upper			101 Ma
	Cenozoic	Paleogene	Paleocene			66 Ma
			Eocene			56 Ma
			Oligocene			33.9 Ma
		Neogene	Miocene			23 Ma
			Pliocene			5.33 Ma
		Quaternary	Pleistocene			2.58 Ma
			Holocene			11.7 ka